

A-OSP Technical Due Diligence

Posture As-Is and Posture Candidate — D1/EQL Proof-Seam Closure (MVP)

CTO / engineering dossier aligned with the A-OSP architecture/topology whitepaper

Field	Value
Document class	Technical due-diligence dossier with embedded architecture specification, runtime evidence, closure gates and engineering readiness plan.
Audience	CTO, staff engineers, enterprise architecture, investor technical diligence, risk/governance, development team.
Version	v.1 live — fresh reconstruction from current main evidence; no reliance on obsolete anchors.
Live anchor	Luke883i/aosp1 main @ e492290b128eb89af44a488d8600d9862f30db0b.
Cross-reference	Full architecture/topology reference: (ii) Augmented Ontological Semantic Platform (A-OSP) Whitepaper - Webapp, Infrastructure, Runtime, Topology.pdf
Non-goals	No legal certification, production certification, complete proof-grade claim, full Oracle enforcement, or benchmarked economics.

Executive posture

Posture as-is: A-OSP1 is no longer merely a contract-only D1 proof concept. Current main contains a dedicated D1 proof route, a D1ProofAppendService, persisted d1_receipt_ semantics, read-after-write evidence, proofSessionContext return and negative tests.

Posture candidate: A-OSP1 is a credible enterprise MVP candidate for proof-sensitive AI-assisted work if the claim remains bounded to D1/EQL proof-seam closure and if C1 artefacts, EQL proof receipts, unified EQL proof routing, downstream gates and pilot evidence pass.

As-is / candidate boundary: D1 has materially advanced on main. EQL proof remains the primary open seam. Full enterprise pilot posture requires P0/P1 gates; production or certification language remains out of scope.

Area	Current posture	External claim boundary
Canonical architecture	AS-IS	Four-layer, text-first SSOT and cache boundaries are established.
D1 proof seam	RUNTIME-WIRED / HARDENING	Route, service, receipt persistence, readback and tests exist; C1 still must prove no bypass.
EQL proof seam	TARGET / OPEN	Contract exists, but no inspected runtime EQLQueryReceipt or unified proof route comparable to D1.
Enterprise readiness	CANDIDATE	Requires P0 proof seam, P1 pilot controls and measured evidence.
Production / certification	NON-GOAL	Do not claim without separate certification/production readiness programme.

1. Audit method and confidence gate

This edition applies an audit discipline designed to reach high confidence without overclaim: every AS-IS statement must have file/line evidence or a generated artefact; every proof claim must have a failure/negative path; every TARGET claim must remain visibly target; every issue reference is roadmap ownership, not delivered code.

Audit question	Pass condition	Current standing
Freshness	Live anchor on current main; no obsolete anchor language.	PASS. This dossier starts from main@e492290b.
D1 runtime evidence	Route + service + server mount + tests.	PASS WITH BOUNDARY. All four are evidenced; C1 still open.
EQL proof evidence	EQLQueryReceipt + unified proof route + tests.	OPEN. Not found in inspected runtime evidence.
Evidence bundle	Human-readable evidence register + hyperlinks.	PASS. Embedded in Section 12 and Appendix A.
Issue catalogue	No 93 vs 100 mismatch; snapshot count is explicit.	PASS WITH TRACKER CAVEAT. Appendix B enumerates 100 rows; regenerate if tracker changes.
CTO usability	Clear decision posture, risk, gates, DoD.	PASS. P0/P1/P2 gates and checklist included.
Engineering usability	APIs, workflow, runtime matrix, failure paths.	PASS. D1 live path and EQL target path are separated.

Audit confidence statement: Based on the inspected repository surfaces and the separation between AS-IS, HARDENING, TARGET and HYPOTHESIS, this dossier is robust enough for CTO / engineering review. It is not a certification artefact. Its strongest claim is bounded: proceed to a scoped D1/EQL closure sprint and pilot readiness assessment.

2. Alignment with “Augmented Ontological Semantic Platform (A-OSP) Whitepaper - Webapp, Infrastructure, Runtime, Topology.pdf”

The file (ii) Augmented Ontological Semantic Platform (A-OSP) Whitepaper - Webapp, Infrastructure, Runtime, Topology.pdf is the architecture/topology whitepaper. This dossier is the live implementation due-diligence layer over aosp1. It narrows the full A-OSP topology to the D1/EQL MVP proof seam while remaining standalone: all key terms, status tags, workflow paths, APIs and gates are restated locally.

Concept in (ii)	Live dossier coordinate	Current posture on main
Browser-native runtime	Canonical stack	AS-IS architecture: React/Vite workspace as projection layer.
Text-first SSOT	.aosp.txt / source-cache taxonomy	AS-IS. Durable truth is text stripes; DB/cache/UI/model/export are not truth.
EQL / FAE / MONL	Query, orchestration and operator layers	EQL proof remains TARGET; FAE/MONL are contextual unless consumed by proof gates.
Error Truth	D1 fail-closed route	PARTIAL AS-IS. Proof D1 route uses RFC 7807 ProblemDetail and explicit FAIL receipt.
Authority Matrix	Evidence discipline	AS-IS concept for resolving doc/generated/code conflicts.
Proof contracts / Proof Spine	D1/EQL closure and downstream composition	D1 advanced; EQLQueryReceipt and Proof Spine remain open/downstream.

3. Terms, status and forbidden equivalences

Term	Meaning
Stripe / .aosp.txt	Append-only text record; durable domain knowledge lives here.
SSOT	Single source of truth; in A-OSP this is the stripe layer.
Cache	Directus/PostgreSQL mirror, index or admin layer; rebuildable, not truth.
D1-D5	D1 intake, D2 enrichment, D3 links, D4 scores, D5 kernels.
Receipt	Typed evidence object emitted by proof-sensitive action.
Readback	Read-after-write verification with canonical hash comparison.
Proof seam	D1 + EQL boundary requiring receipts, readback/scope and negative tests.
EQL	A-OSP DSL for stripe-shaped, scoped data; not generic SQL.
FAE	Focus/Attention/Execution boundary for model calls.
MONL	Operator layer over parsed stripes.
Error Truth	Typed failure semantics; failure must not become false proof.
Proof Spine	Target chain of receipts across intake, retrieval, enrichment, artifact and review.
Implementation status	Use
AS-IS	Evidence exists at live anchor.
RUNTIME-WIRED	Live runtime path is mounted or invoked.
HARDENING	Partially implemented and being strengthened.
CONTRACT-ONLY	Schema/helper exists; does not prove runtime.
TARGET	Issue-owned or planned; not delivered.
HYPOTHESIS	Measurement needed before claim.
NON-GOAL	Explicitly outside current scope.
Forbidden equivalence	Replacement rule
fluent output = proof	Only receipt/readback/scope/witness can support proof.
confidence = evidence	Confidence is not source evidence.
log = receipt	Logs diagnose; receipts carry proof semantics.
export = witness	Only receipt-backed export/witness can be relied upon.
UI green = proof	Only valid receipt-derived UI state may be green.
fallback/demo/local = proof	Must fail closed.
model memory = SSOT	Provider is replaceable processor.
raw query success = proof	Requires explicit scope and EQLQueryReceipt.

4. Canonical stack and repository geometry

A-OSP v3 is defined as a 4-layer architecture: CORE, GENERATOR, ADAPTER and GOVERNANCE, with text files as the single source of truth, database as queryable cache, and a pipeline from unstructured knowledge to auditable artifacts. ARCHITECTURE.md is canonical at the live anchor [E01].

Layer	Responsibility	Representative technology / surface
CORE	Data models, stripe files, D1-D5, kernel, schema validation, parser/MONL.	TypeScript, Zustand, Node.js, Express 5, Zod, PostgreSQL, Redis.
GENERATOR	UI apps, backend services, CLI, OS-like desktop and visualisations.	React 19, Vite 7, Ant Design, Monaco.
ADAPTER	Third-party wrappers, Directus bridge, legacy interop, proxy configs.	Directus, Traefik, Docker.
GOVERNANCE	CI/CD, docs, authority matrix, generated artefacts, agent governance.	GitHub Actions, governance scripts, iKant-governance.

Surface	Canonical status	Rule
.aosp.txt	Durable source / SSOT	All domain data lives in append-only text stripes.
Parser-service	Custom API + Stripe Engine	Primary backend; not a thin no-code wrapper.
Directus/PostgreSQL	Queryable cache / admin / auth	Mirror/cache, not source of truth.
React UI	Projection / workspace	Renders proof state; cannot manufacture proof.
LLM provider	Replaceable processor	Bounded calls; no durable truth.
PDF/DOCX/JSON export	Downstream representation	Not proof unless backed by witness/receipt.

Repository geometry note: docs/audit/STACK_LAYER_CENSUS.md reports 1,697 frontend source files, 500 test files and 20 registered apps, but the file itself is marked AI-generated / NON-SSOT and has an older dynamic refresh. Treat those counts as advisory inventory, not a fresh measurement. Fresh release circulation should regenerate the census.

5. Build, dependency and guard surface

Frontend package evidence shows Node >=20, npm >=10, Vite build, Vitest, Playwright, golden D1 E2E, storage-SSOT lint, no-D1-fire-and-forget lint and no-deprecated-D1-paths lint. These are important because they show the repo has guard surfaces, not merely product code [E08].

Root package evidence exposes audit:d1-dod, audit:eql, proof, proof-spine, proof-readback, governance fake-green checks, forbidden-equivalence lint, OpenAPI validation, topology scans and CI topology checks [E09].

Evidence gate	Status	Next action
Bundle size	UNKNOWN	Generate real bundle report before external performance claims.
Dependency risk	PARTIAL	Generate dependency tree and zombie-dependency report.
Latency / proof overhead	UNKNOWN	Measure append, readback and EQL p50/p95/p99.
Security posture	PARTIAL	Keep CORS/secrets/auth evidence separate from proof claims.

6. Proof architecture

6.1 D1 proof route

Current main contains backend/parser-service/src/routes/proof-d1.ts. The route header states it implements POST /api/v1/proof/d1/append, is separate from generic append and /api/v1/d1/* secondary paths, and returns receipt plus proofSessionContext on success. It also documents RFC 7807 ProblemDetail failures and explicit FAIL receipt on non-convergence [E02].

```
POST /api/v1/proof/d1/append
Success: { ok, receipt, proofSessionContext, written }
Failure: application/problem+json; 422 carries explicit FAIL receipt when readback fails
```

6.2 D1ProofAppendService

The service declares itself the single proof-grade producer of D1 receipts. It writes answer_ -> atom_ -> atom_validation_ -> audit_ -> d1_receipt_, and states the separation rule: generic append may create canonical data, but only this seam can produce proof PASS after read-after-write [E03].

```
answer_ -> atom_ -> atom_validation_ -> audit_ -> d1_receipt_
PASS only after read-after-write confirms canonical atom stripe
```

Runtime detail: the service writes canonical stripes, executes a bounded read-after-write loop, computes a canonical hash, persists d1_receipt_, and returns proofSessionContext with sessionId, receiptId, readbackHash, mode and status [E04].

6.3 D1 tests

backend/parser-service/test/proof-d1.test.js validates the proof seam: happy path persists d1_receipt_, PASS carries readback, idempotency token is deterministic, failed readback yields FAIL and never PASS, append error yields FAIL, stale/tampered atom mismatch does not converge [E05].

6.4 EQL proof boundary

EQL remains the primary open seam. The contract defines EQL as an A-OSP DSL, not small SQL. EQL-Proof requires explicit scope and receipt semantics, but the inspected runtime search for EQLRunResult / EQLQueryReceipt / eqlQueryService did not expose a comparable runtime proof implementation. Issue #3488 still defines EQL proof as EQLQueryReceipt + queryHash + resultHash + explicit scope [E06, E07].

7. End-to-end workflows

7.1 D1 live path

Step	Runtime status	Mechanism
1. User completes D1 intake	Workflow context	DataIntake generates answer/atom/validation payload.
2. UI calls proof route	RUNTIME target must use route	POST /api/v1/proof/d1/append, not generic append.
3. Route validates body + permission	RUNTIME-WIRED	zod schema + RBAC middleware.
4. Service writes canonical stripes	RUNTIME-WIRED	answer_, atom_, atom_validation_, audit_.
5. Readback barrier	RUNTIME-WIRED	readSession, parseFileContent, find atom_, hash compare.
6. Receipt persistence	RUNTIME-WIRED	append d1_receipt_ with status/hash/readback fields.
7. proofSessionContext	RUNTIME-WIRED	sessionId, receiptId, readbackHash, mode, status returned.
8. UI/D2 consumers	HARDENING / must verify	Must consume receipt, not completion/progress.

7.2 D1 failure paths

Failure	Expected behavior	Evidence / gate
generic append success	canonical data, not proof PASS	Route comments explicitly separate generic append from proof path.
storage unavailable	503 ProblemDetail	proof-d1 route fail-closed behavior.
readback mismatch / non-convergence	422 ProblemDetail + FAIL receipt	route and service behavior.
append error	FAIL receipt, never PASS	proof-d1 tests.
stale/tampered atom	FAIL; no convergence	proof-d1 stale/tamper test.
D2 without receipt	must fail	C1 / #3488 required negative test, not fully closed until C1.

7.3 EQL target path

Step	Expected EQL-Proof behavior	Status
1. Surface declares EQL level	Core / Proof / Analytic	Contract / target.
2. User submits PROOF SELECT	Must include IN SESSION	Contract / target.
3. Strict parser builds AST	No silent fallback	Engine foundations cited as landed in docs; runtime proof remains target.
4. Scope canonicalizer	Reject implicit/global proof scope	Target negative test.
5. Executor reads stripes/views	_stripe_, _receipt_, _phase_, _lineage_	Target proof view semantics.
6. Result classified	ROWS / VALID_ZERO / FAIL_EMPTY / FAIL_SCOPE / FAIL_PARSE	Target.
7. Receipt persisted	EQLQueryReceipt / eql_query_	MISSING / TARGET in inspected runtime evidence.
8. Downstream consumes receipt	ArtifactWitness / Proof Spine	Target / downstream.

8. API status

API	Current posture	Meaning
POST /api/v1/proof/d1/append	RUNTIME-WIRED / HARDENING	Dedicated D1 proof append route.
POST /api/v1/sessions/:id/append	CURRENT generic append	May create canonical data; must not create proof PASS.
GET /api/v1/d1/capability	RUNTIME-WIRED	Mounted under /api/v1/d1/capability; capability surface.
POST /api/v1/query	CURRENT query	Generic query route; not automatically EQL-Proof.
GET /api/v1/monl/status	CURRENT MONL route family	Operator/status route, not proof by itself.
POST /api/eql/run	TARGET / not found as runtime in inspected evidence	Expected unified EQL proof route.

9. Implementation state matrix

Surface	Live posture	Evidence	Open risk
4-layer architecture	AS-IS	ARCHITECTURE.md [E01]	Low.
.aosp.txt SSOT / Directus cache	AS-IS	ARCHITECTURE.md backend boundaries [E01]	Medium if app bypasses filesystem.
D1 proof route	RUNTIME-WIRED	proof-d1.ts [E02], server mount [E10]	Consumer adoption must be confirmed.
D1ProofAppendService	RUNTIME-WIRED	d1-proof-append-service.ts [E03,E04]	Contract mirror with shared proof types should be kept in sync.

d1_receipt_persistence	RUNTIME-WIRED	persistReceipt [E04]	Receipt schema consistency and EQL view bridge.
D1 negative tests	LANDED	proof-d1.test.js [E05]	CI inclusion must be confirmed.
D2 hard gate	TARGET / must verify	C1/#3488	Potential bypass until proven.
EQLQueryReceipt	TARGET / not runtime-evidenced	search + #3488 [E06,E07]	Primary remaining proof seam.
C1 artefacts	TARGET	Issue #3488 required outputs [E07]	Gate not closed.
Bundle/perf metrics	UNKNOWN	No generated release bundle artefact in inspected evidence	Generate before external performance claims.

10. Gap register and closure path

Gap	Current status	Owner / gate
D1 route/service/receipt/readback	Substantially implemented on main	Still validate no legacy bypass via C1.
D1 consumer adoption	HARDENING	All UI/D2/Finder/Shell consumers must use proofSessionContext.
D2 hard gate	TARGET / verify	No D2 without D1Receipt PASS + readback.
d1_receipt_EQL bridge	TARGET / verify	_receipt_virtual view must include D1 receipts.
EQLRunResult / EQLQueryReceipt	TARGET	Issue #3425 / #3488.
Unified EQL proof route	TARGET	Issue #3426 / #3488.
EQL canonical hashes	TARGET	Issue #3427 / #3488.
EQL proof CI	TARGET	Issue #3430 / #3488.
C1 artefacts	TARGET	docs/generated report + scans + false-green results.

Closure order: D1 consumer adoption and D2 hard gate must be verified in parallel with EQL receipt implementation. The enterprise MVP posture is not strengthened by adding downstream features before C1 closes.

11. Metrics and Definition of Done

P0 metric	Target
d1_receipt_presence_rate	100% for proof D1 actions.
d1_pass_without_readback_count	0.
d1_fail_receipt_on_error_rate	100% for storage/readback failures.
d2_handoff_without_d1receipt_count	0.
eql_proof_without_scope_count	0.
eql_success_without_receipt_count	0.
generic_append_to_proof_pass_count	0.
bundle_size_report_present	100% per release candidate.
Gate	Required before claim
P0 proof seam	D1 route/service tests pass; D2 gate verified; EQLQueryReceipt + unified proof route + EQL negative tests implemented; C1 artefacts generated.
P1 enterprise pilot	Security posture, recovery, observability, provider exposure, authority matrix, perf/bundle reports and runbooks complete.
P2 scale readiness	Concurrent proof-mode workload, tenant isolation tests, cost model, failover, ArtifactWitness/Proof Spine consumption.

12. Evidence register

ID	Source	Why it matters	Link
E01	ARCHITECTURE.md#L3-L8, #L24-L30, #L34-L42, #L46-L72, #L88-L117	Canonical architecture, authority matrix, principles, layers, backend boundaries.	open
E02	backend/parser-service/src/routes/proof-d1.ts#L3-L24, #L89-L160	D1 proof append route, request validation, fail-closed ProblemDetail.	open
E03	backend/parser-service/src/services/d1-proof-append-service.ts#L3-L24, #L196-L234	D1ProofAppendService as single proof producer; request/result types.	open
E04	d1-proof-append-service.ts#L236-L268	Writes canonical stripes, readback loop, d1_receipt_persistence, FAIL receipt.	open
E05	backend/parser-service/test/proof-d1.test.js#L51-L179	D1 happy path and false-green / stale / append-error tests.	open
E06	docs/EQL_MIGRATION_PLAN.md and search results	Search for EQLRunResult/EQLQueryReceipt returned docs/plans, not comparable runtime implementation.	open
E07	Issue #3488	C1 D1/EQL proof seam, required artefacts and negative tests.	open
E08	frontend/package.json#L9-L83	Frontend scripts and guard surface.	open
E09	package.json#L31-L100	Root proof/governance/audit scripts.	open
E10	backend/parser-service/src/server.ts#L122-L135	D1 workflow API, D1 capability and proof D1 route mounts.	open

13. CTO and engineering checklist

Question	Expected answer before pilot
Where is truth?	.aosp.txt stripes are SSOT; DB/cache/UI/model/export are not truth.
Can UI green fake proof?	No; UI must derive proof state from valid receipt and fail closed.
Can generic append create proof PASS?	No; only D1 proof route/service may produce PASS receipt.
Can D2 consume unproven D1?	No; D2 handoff must require D1Receipt PASS + readback.
Can raw EQL success be proof?	No; EQL proof requires explicit scope and EQLQueryReceipt.
Are issue references treated honestly?	Yes; issues are TARGET/ownership evidence, not delivered code.
Are bundle/perf economics measured?	Not yet; no quote until artefacts are generated.
Is the cross-reference to (ii) clear?	Yes; (ii) is architecture/topology; this is live implementation due diligence.

14. Final posture statement

As-is posture: A-OSP1 main contains a coherent canonical architecture, substantial proof contracts, a dedicated D1 proof route, a D1ProofAppendService, d1_receipt_persistence, read-after-write evidence and D1 negative tests.

Candidate posture: A-OSP1 is credible as an enterprise MVP candidate for proof-sensitive AI-assisted work only if it keeps the claim bounded to the D1/EQL proof seam and completes EQL receipt/routing, C1 artefacts, downstream gates and measured pilot evidence.

Recommended next action: proceed to a scoped proof-seam closure sprint. Do not use production readiness, compliance certification or complete proof-grade language until P0/P1 gates pass.

Appendix A — Resolved evidence hyperlinks

All links below are commit-pinned to main@e492290b128eb89af44a488d8600d9862f30db0b. They require repository access if the repository is private.

E01 — ARCHITECTURE.md#L3-L8, #L24-L30, #L34-L42, #L46-L72, #L88-L117: Canonical architecture, authority matrix, principles, layers, backend boundaries. [open source](#)

E02 — backend/parser-service/src/routes/proof-d1.ts#L3-L24, #L89-L160: D1 proof append route, request validation, fail-closed ProblemDetail. [open source](#)

E03 — backend/parser-service/src/services/d1-proof-append-service.ts#L3-L24, #L196-L234: D1ProofAppendService as single proof producer; request/result types. [open source](#)

E04 — d1-proof-append-service.ts#L236-L268: Writes canonical stripes, readback loop, d1_receipt_ persistence, FAIL receipt. [open source](#)

E05 — backend/parser-service/test/proof-d1.test.js#L51-L179: D1 happy path and false-green / stale / append-error tests. [open source](#)

E06 — docs/EQL_MIGRATION_PLAN.md and search results: Search for EQLRunResult/EQLQueryReceipt returned docs/plans, not comparable runtime implementation. [open source](#)

E07 — Issue #3488: C1 D1/EQL proof seam, required artefacts and negative tests. [open source](#)

E08 — frontend/package.json#L9-L83: Frontend scripts and guard surface. [open source](#)

E09 — package.json#L31-L100: Root proof/governance/audit scripts. [open source](#)

E10 — backend/parser-service/src/server.ts#L122-L135: D1 workflow API, D1 capability and proof D1 route mounts. [open source](#)

Appendix B — Open issue snapshot (roadmap evidence)

#	Area	Where it intervenes	What it does
#3037	Parser-service / budget	backend/parser-service, OpenAPI, endpoint budget	Classifies and caps parser-service endpoint growth; prevents API sprawl.
#3038	CI topology	.github/workflows, required/advisory checks	Maps and rationalises CI workflows, distinguishing blocking, advisory, scheduled, manual and archive.
#3040	E2E contracts	Playwright / critical journeys	Turns E2E coverage from file-exists into journey-executed runtime proof.
#3041	iKant E2E deferred	Playwright iKant specs	Schedules or correctly defers 6 iKant specs; avoids smoke-test false green.
#3042	Critical journeys	PR-blocking E2E	Defines PR-blocking critical journeys, including D1/D2/Artifact proof paths.
#3044	Live-stack proof matrix	local / PR / nightly / provider-real	Defines real/mock backend, mock/real provider, seed and trigger matrix.
#3045	D1-D5 canonical path	frontend pipeline, hooks, backend pipeline	Canonicalises one D1-D5 path and tombstones alternate/legacy paths.
#3046	Side-effect scanner	AST scanner, timers, singletons, globals	Audits setInterval, module-level singleton allocation and unsafe globals.
#3047	Control plane	aosp.sh, npm scripts, docker/devcontainer	Elects/simplifies startup/verify/reset path; archives dormant scripts.
#3048	Support truth	active/dormant/dead/archaeological registry	Reconciles support status of surfaces into one truth.
#3050	Classification registry	CLASSIFICATION_REGISTRY, generated/advisory docs	Makes registry derived, verified or explicitly advisory.
#3051	Runtime ontology docs	README, ARCHITECTURE, DecisionLog	Rewrites active docs on runtime ontology; removes semantic legacy.
#3052	Authority conflicts	.ikant, .github, blueprints, vision docs	Resolves blueprint/document authority conflicts and high-visibility stale docs.
#3053	Legacy purge	dead/dormant semantic leftovers	Tombstones/removes dead or valueless dormant surfaces; archives design value.
#3054	Semantic anti-drift	authority matrix, generated docs, topology	Anti-drift pipeline for docs, topology, inventory and E2E reports.
#3056	Agent-layer authority	.agent/, .ai/, meta-governance	Classifies agent/AI directories by tier, owner and write access.
#3077	Perimeter map	moat/custom/substrate/delete map	Classifies every surface as moat-custom, substrate-standard or to-delete.
#3078	Hybrid architecture ADR	DecisionLog / ADR	Formalises hybrid architecture: custom moat + standard substrate.
#3079	FS SDK	useFilesystem, atom/strip SDK	Extracts filesystem abstraction into @aosp/fs-sdk.
#3080	App SDK	app manifest, lifecycle, sandbox	Publishes @aosp/app-sdk for app manifests, lifecycle hooks and sandbox boundary.
#3081	Enterprise auth	Keycloak / SSO	Replaces custom JWT with Keycloak SAML/OIDC/SCIM.
#3082	Authorization	Cerbos / ABAC / SoD	Replaces custom RBAC with ABAC/SoD policy engine.
#3083	Audit WORM	audit trail, Postgres hash chain	Makes audit trail append-only and hash-chained.
#3084	Evidence store	MinIO / Object Lock	Introduces WORM-compatible evidence storage with S3 Object Lock.
#3085	Workflow engine	Camunda BPMN	Integrates human-in-loop GRC approval workflow.
#3086	Secrets	Vault	Replaces .env/SOPS with HashiCorp Vault.
#3087	Observability	OTel, Grafana, traces/logs/metrics	Standardises monitoring and incident response on OTel/Grafana.
#3088	Substrate matrix	compatibility / version pinning / SLA	Generates substrate compatibility matrix and adapter SLAs.
#3089	Adapter tests	substrate adapters	Adds contract, integration and failure-mode tests for substrate adapters.
#3090	GRC mapping moat	MONL, compliance frameworks	Multi-framework mapping engine for GDPR/NIS2/ISO/SOC2/DORA/AI Act.
#3091	Control library	seed controls	Seeds about 500 pre-authored compliance controls.
#3092	Risk quantification	FAIR-lite / Monte Carlo	Quantitative-risk module with simulations.
#3093	Regulatory feed	reg-tech / official RSS feeds	Regulatory horizon scanning via official or provider feeds.
#3094	E-signature	evidence lifecycle / 21 CFR / eIDAS	Evidence signatures with re-auth, hash-chain and optional QES provider.
#3095	Multi-tenant isolation	DB schema, bucket, realm	Hard tenant isolation: DB schema, MinIO bucket, Keycloak realm.
#3096	Data residency	EU/US/APAC deploy templates	Deployment templates for data residency.
#3100	App store	registry, signing, review queue	Production app store for third-party apps with signing and sandbox hardening.
#3102	GRC enterprise epic	12-month Scenario C plan	Enterprise GRC readiness epic coordinating substrate, moat and timeline.
#3392	Governance base	governance contract / inventory	Single Governance Contract + total inventory; real vs advisory checks.
#3393	Governance orchestrator	PR/governance orchestration	Centralised governance orchestrator for assessment outputs, checks and summaries.
#3394	PR Bot Summary	PR-facing governance	Makes PR Bot Summary the single PR-facing summary.
#3395	Typed docs	docs as typed nodes	Types docs as canonical, operational, generated, advisory, archive.
#3396	Governance cutover	fake/legacy surfaces	Final cutover to governance framework; removes fake/legacy surfaces.
#3420	EQL meta	EQL v2 program	EQL tracker, DAG, ADR and hard gates.
#3424	EQL virtual views	__stripe__, __receipt__, __phase__, __lineage__	Implements proof-aware virtual views.
#3425	EQL receipt schema	EQLRunResult, EQLQueryReceipt, service stub	Introduces envelope/schema for EQL results and receipts.
#3426	EQL route unification	parser-service EQL routes	Routes all EQL through eqLQueryService, with no bypass.
#3427	EQL hash/receipt	canonicalisation, eqL_query_	Computes deterministic hashes and writes EQL receipts.
#3428	EQL UX	Shell/Finder/Workstation	Deterministic UX: parse, support, scope, valid-zero/error and proof status.
#3429	EQL analytic/lineage	analyze / trace	Adds analytics and lineage without SQL sprawl.
#3430	EQL final convergence	proof scripts, fixtures, CI	Closes EQL v2 with proof runner, fixtures, chaos, CI and cleanup.
#3431	EQL anti-goals	no SQL sprawl, no global proof	Hardens anti-goals: no full SQL, no implicit global proof query.
#3434	Oracle M9	prompt/config registry	Versioned iKant/Oracle prompts editable from UI.
#3435	Oracle M10	transition fabric	Traceable cross-app iKant/Oracle transitions.
#3436	Oracle M11	governance workbench	Advanced workbench for profile/observer/kernel/policy.
#3437	Oracle M12	capstone E2E	Oracle capstone: E2E, UX polish, cleanup, propagation and proof pack.
#3445	Error Truth E1	AOSPErrorEnvelope	Canonical error envelope and lossless projections.
#3446	Error Truth E2	request attribution	Deterministic UI to backend to Observatory to audit attribution.
#3447	Error Truth E3	app adoption	Migrates apps to Error-Truth contract.
#3448	Error Truth E4	UI language	User-facing errors carry status, cause, scope, proof and recovery.
#3449	Error Truth E5	forensic exports	Forensic-grade ErrorHistory, Audit Trail, Dashboard and /errors.
#3450	Error Truth E6	AI clustering	AI-assisted but deterministic-first error clustering.
#3451	Error Truth E7	local hardening	Local hardening: voice, LLM config, Observatory, polling/backoff.
#3462	D1 proof runtime	DataIntake / D1 proof seam	D1 proof runtime: append/readback and proof-grade DataIntake seam.
#3465	Oracle capstone audit	post-M12 hardening	Audits Oracle Saga capstone and post-M12 plan.
#3466	Oracle M8ter	UI/UX exposure baseline	Oracle UI/UX baseline and anti-false-green hardening.

#3470	DoC / Living Contracts	Milestone Ledger / MVP chain	Governance control tower for convergence and proof chain.
#3471	D2-D5 receipts	PhaseReceipt / PipelineReceipt	Receipts and lineage for D2-D5.
#3472	Artifact Witness	ArtifactWorkstation / witness	ArtifactWitness MVP; export is not witness.
#3473	Proof Spine	MVPProofBundle	Proof spine validator fails honestly when artefacts are missing.
#3483	UI grammar	product-facing labels/layout/copy	Canonical UI grammar preventing copy/layout from faking proof.
#3484	MONL-TX	transaction membrane	MONL proof-aware transaction engine: event accepted is not committed.
#3485	Entropy scaffold	deterministic convergence	Saga map, entropy rules, agent contract and false-equivalence detection.
#3486	MVP readiness backlog	canonical issue map	Ordered map of non-post-MVP issues; monitors proof sequence.
#3487	C0 convergence	corpus authority / cleanup preflight	Authority, legacy discovery and generated-docs policy before runtime proof.
#3488	C1 convergence	D1/EQL proof seam	D1/EQL checkpoint: D1Receipt, readback, proofSessionContext, EQL receipt/scope.
#3489	C2 convergence	D2-D5 receipt chain	Converges D2-D5 with receipts/readback/lineage.
#3490	C3 convergence	ArtifactWitness / Proof Spine / Governance	Final proof composition: no UI/export/progress as proof.
#3491	C4 convergence	UI / Error Truth / UX sanitation	UI, errors, labels and empty states must not appear as proof.
#3492	C5 convergence	MONL / perimeter / substrate	Conditional checkpoint over MONL, perimeter and substrate quarantine.
#3493	Agent drift control	governance grammar	Harmonises Entropy, Convergence, agent governance, CI/CD and PR bots.
#3542	D1 vertical slice	D1Receipt E2E loop	Minimal input->append->persist->readback->consume loop.
#3543	D1 receipt adapter	persist/readback + useD1ProofStatus	Emits d1_receipt_ and makes it readable by proof-status hook.
#3544	CanonicalAppendSDK wire	D1 append proof-grade	Routes every proof-grade D1 append through canonical SDK.
#3545	Strict proof mode	fallback/mock/demo fail-closed	Blocks fallback, mock and demo from counting as proof.
#3546	CompletionModal gate	D1->D2 UI gate	No D2 path unless D1Receipt is PASS.
#3547	proofSessionContext	D1->EQL->D2 context	Propagates traceable context ID across D1, EQL and D2.
#3548	D1 receipt EQL bridge	_receipt_, EQLQueryReceipt	Makes d1_receipt_queryable and connected to EQL scoped receipt.
#3549	D1->D2 barrier	/handoff	Hard-gates D2 handoff on D1Receipt PASS.
#3550	D1 attack tests	13 false-green vectors	Automated fail-closed tests against false-green vectors.
#3551	D1 validated != proof	validated=true regression	Ensures validated=true never becomes proof without receipt.
#3552	Fixture/runtime separation	CI proof artefacts	Separates runtime receipts from fixtures; no fixture-as-proof.
#3553	D1 golden E2E	PR-blocking proof journey	Golden runtime journey with real proof artefacts.
#3554	Lint governance	no-duplicate-scroll	Stale lint rule: promote, delete or formally extend.
#3555	Lint governance	no-cursor-text-on-containers	Governance for stale lint rule older than 60 days.
#3558	D1 follow-up meta	D1-FU.1..10 retirement	Epic to retire documented-but-unproven D1 self-debt.
#3560	D1FU-B tests	skip->fail + DI	No defensive skips; no test-only global resets.
#3561	D1FU-C CI guards	guard convergence	Converges lint/guard CI: emoji lint, D-030 dedupe, guards, KC fixed-point.
#3562	D1FU-D governance SoT	role-bypass ADR + one-pager	Moves bypass and one-pager runnability into authoritative sources.
#3580	CI nondeterminism	Vitest forks teardown race	Fixes nondeterministic Vitest assessment outputs with JSON reporter/sharding/OOM.